

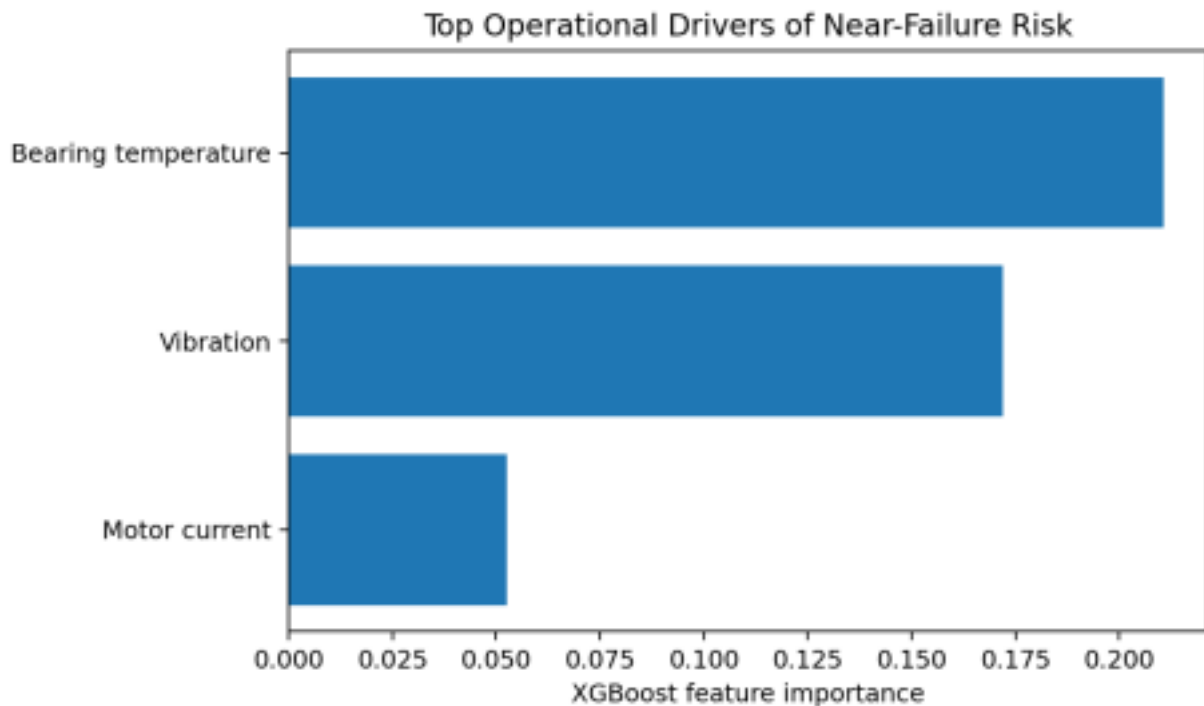
Week 9 Model Explainer — Predictive Maintenance

The Analogy

The model can be understood as a digital version of an experienced maintenance supervisor walking through a pump station. The supervisor does not decide that a pump is healthy or dangerous by looking at one gauge. Instead, the supervisor notices patterns: the pump is vibrating more than usual, its bearing temperature is rising, the motor is drawing unusual current, and maintenance has been overdue. Each sign alone may not justify shutting down the pump. Together, however, they can indicate that the pump deserves attention.

The machine-learning model performs a similar task. It combines several operational signals and assigns a probability that the pump is in a near-failure condition. The purpose is not to replace engineers. Its purpose is to help the operations team decide where to look first.

The Drivers



- **Bearing temperature:** Higher bearing temperature was one of the strongest signals in the model. In operational terms, excessive heat can indicate increased friction, lubrication problems, overload, or other developing mechanical stress. A rising temperature trend should therefore prompt inspection rather than being treated as an isolated alarm.
- **Vibration:** Vibration was the strongest global SHAP driver. The model frequently associated elevated vibration with higher near-failure risk. This is operationally intuitive because abnormal vibration can accompany imbalance, misalignment, bearing degradation, looseness, or other mechanical problems.
- **Motor current:** Motor current added information about the electrical and mechanical load on the pump. A change in current can become more meaningful when it occurs together with abnormal vibration or temperature. The key insight is therefore the combination of warning signals rather than one sensor crossing a universal threshold.

Trust and Limits

The model is intentionally evaluated with Precision, Recall, F1-score, ROC-AUC and the Confusion Matrix rather than accuracy alone. For predictive maintenance, a high accuracy score can be

misleading if failure events are uncommon. A model that predicts 'no failure' most of the time may appear accurate while still missing the cases that matter operationally.

False negatives:

A false negative occurs when the model does not flag a pump that is genuinely approaching failure. This is the more safety-sensitive error because the maintenance team may lose an opportunity to inspect or intervene. For that reason, the capstone target prioritises at least 80% recall.

False positives:

A false positive occurs when the model flags a pump that does not subsequently experience a near-failure event. False positives are not harmless: they can consume technician time, trigger unnecessary inspections, and eventually create alert fatigue. The capstone therefore also targets a false-positive rate below 5%.

How operations should use the tool:

The model should be used as a prioritisation and decision-support layer. A high-risk score should trigger a maintenance review, followed by verification of sensor quality, recent maintenance history, process conditions, engineering limits, and applicable safety procedures. The final maintenance decision remains with qualified personnel.

Important limitation:

The primary Week 9 dataset is physics-informed synthetic SCADA data because real KPC internal pump and maintenance logs are not publicly available. The resulting metrics therefore demonstrate the ML workflow and operational trade-offs, not proven production performance. Before deployment, the approach must be validated on real operational data and monitored for sensor drift, missing data, changing operating regimes, and degradation of model performance.

Operational conclusion

The model's value is not that it can predict the future with certainty. Its value is that it can help a limited maintenance team focus attention on the pumps whose combined telemetry most resembles previously learned high-risk patterns. Used transparently and alongside engineering judgment, this supports the capstone's broader transition from calendar-based maintenance toward condition-based maintenance.